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Degree: B.Tech

Roll no.:
Semester: 3

Mid Semester Examination, September 2024

Course Code: EIMTC04/ITMTC04/INMTC04
Course title: Optimization principles and techniques

Time: 1.5 hours

Maximum Marks. 25

Note: Missing data/information(if any), may be suitably assumed and mentioned in the answer.

Q. No.	Questions	Marks	CO
1a	An electronic company is engaged in the production of two components C_1 and C_2 used in the TV sets. Each unit of C_1 costs the company Rs 25 in wages and Rs 25 in material, while each unit of C_2 costs Rs 125 in wages and Rs 75 in material. The selling price of C_1 is Rs 150 per unit and of C_2 is Rs 350 per unit. The company's production capacity is limited by two considerations. The company has initial balance as Rs 20,000. The company has 4000 hours of machine time and 2,800 hours of assembly time. The production of C_1 requires 6 hours of machine time and 4 hours of assembly time, whereas the production of C_2 requires 4 hours of machine time and 6 hours of assembly time. Formulate this problem as LPP so as to maximize total profit to the company.	2.5	CO1
1b	Use graphical method to solve the LPP. Max $Z = 2x_1 + 4x_2$ subject to $x_1 + 2x_2 \leq 5$ $x_1 + x_2 \leq 4$, $x_1, x_2 \geq 0$.	2.5	CO1
2a	Use two-phase simplex method to maximize Max $Z = 5x_1 - 4x_2 + 3x_3$ subject to $2x_1 + x_2 - 6x_3 = 20$ $6x_1 + 5x_2 + 10x_3 \leq 76$ $8x_1 - 3x_2 + 6x_3 \leq 50$ $x_1, x_2, x_3 \geq 0$.	2.5	CO1
2b	Find degenerate basic feasible solutions to the system $x_1 + x_2 + x_3 = 3$ $x_1 - x_2 + x_4 = 0$ $x_1, x_2, x_3, x_4 \geq 0$.	2.5	CO1
3a	State Complimentary Slackness Theorem. Consider the following LPP: Max $Z = 3x_1 + 4x_2$ subject to $x_1 + x_2 \leq 12$ $2x_1 + 3x_2 \leq 30$ $x_1 + 4x_2 \leq 36$, $x_1, x_2 \geq 0$. The solution to the above LPP is $x_1 = 6, x_2 = 6$, and $Z = 42$. Write the dual of the above LPP. Also find the solution of the dual problem using complimentary slackness theorem.	2.5	CO2
3b	Solve the following linear programming problem using dual simplex method. Min $Z = 4x_1 + 7x_2$ subject to $2x_1 + 3x_2 - x_3 = 5$ $x_1 + 7x_2 - x_4 = 9$ $x_1, x_2, x_3, x_4 \geq 0$.	2.5	CO2